

1 Problem Formulation and Main Result

Consider an $n \times n$ grid of numbers where each position (i, j) is the reciprocal of the smaller of the two indices. Explicitly, we define the matrix A_n by

$$a_{ij} = \frac{1}{\min(i, j)} \quad \text{for } 1 \leq i, j \leq n.$$

For instance, when $n = 4$, the matrix takes the explicit form

$$A_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{3} \\ 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \end{pmatrix}.$$

The goal, originally featured on the William Lowell Putnam Mathematical Competition, is to compute the closed form of $\det(A_n)$. While standard evaluations typically rely on elementary row operations, the derivation presented here approaches the problem through discrete calculus by formulating a recurrence relation on consecutive determinant ratios.

When a structure follows a pattern this explicit, our first instinct shouldn't be to throw standard linear algebra tools at it so mechanically. Instead of treating the matrix as just a set of numbers, it helps to view it as a dynamic object obeying a single rule, and ask a simpler question: how does it change from one step to the next?

To begin, let's look at how the values distribute across the grid. The matrix exhibits a visual “flow”: L-shaped rings of constant values cascading downward into the bottom-right corner.

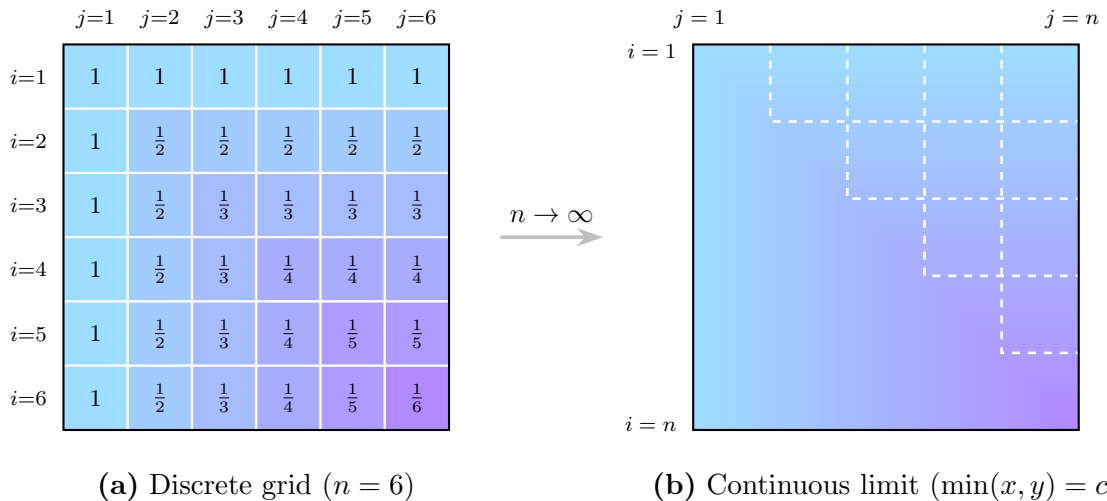


Figure 1: The grid of A_n . In the discrete matrix (a), entries remain constant along concentric L-shaped rings. In the continuous limit (b), these blend into a smooth “flow” of shrinking values toward the bottom-right corner: $\min(x, y) = c$.

At this point, the nature of the change becomes clear: that it's seemingly uniform. This regularity is our primary lead: by defining how each layer changes, we can create a direct path to the closed-form solution.

To turn this uniformity into a usable result, we need to track how the system changes from one step to the next; however, before we do that, we have to choose what scale of change to measure. While several options exist, the most natural candidate is dimensional: studying the ratio between the determinant of each $n \times n$ matrix and the one before it.

Computing the determinants up to $n = 4$ and tracking their successive ratios $r_n = D_n/D_{n-1}$ yields:

$$\begin{aligned}
 D_1 &= \det \left(\begin{bmatrix} 1 \end{bmatrix} \right) = 1 && \text{(base case)} \\
 D_2 &= \det \left(\begin{bmatrix} 1 & 1 \\ 1 & \frac{1}{2} \end{bmatrix} \right) = -\frac{1}{2} && \implies r_2 = \frac{-1/2}{1} = -\frac{1}{2 \cdot 1} \\
 D_3 &= \det \left(\begin{bmatrix} 1 & 1 & 1 \\ 1 & \frac{1}{2} & \frac{1}{2} \\ 1 & \frac{1}{2} & \frac{1}{3} \end{bmatrix} \right) = \frac{1}{12} && \implies r_3 = \frac{1/12}{-1/2} = -\frac{1}{3 \cdot 2} \\
 D_4 &= \det \left(\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{3} \\ 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \end{bmatrix} \right) = -\frac{1}{144} && \implies r_4 = \frac{-1/144}{1/12} = -\frac{1}{4 \cdot 3}
 \end{aligned}$$

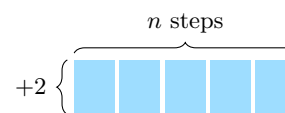
At first glance, the denominators 2, 6, 12, ... might not appear at all special. However, if we track how the rate of change itself evolves, a pattern starts to form: the step sizes increase with regularity, meaning that beneath the fractions lies a constant.

Dimension n :	2	3	4	5	6
Denominator d_n :	2	6	12	20	30
First Difference (Δ) :	+4	+6	+8	+10	
Second Difference (Δ^2) :		+2	+2	+2	

To understand why this constant second difference is so important, we can reverse the table by accumulating the changes from the bottom up. Even without formal calculus, the arithmetic behind this accumulation has a well-defined pattern:

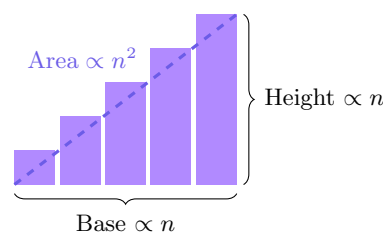
1. Accumulating a constant creates a line.

If every step adds the exact same value (+2), taking n steps means adding 2 a total of n times. Repeated addition is just multiplication ($2 \cdot n$), turning a constant increment into a strictly linear sequence.



2. Accumulating a line creates an area.

If each new step is linearly larger than the last (2, 4, 6, ...), stacking these increments side-by-side forms a discrete triangle. Because both the base and the height grow proportionally with n , the accumulated area scales with $n \times n$, naturally producing a quadratic curve.



Because the second difference of our denominators is constant, the first difference must grow linearly. And as shown above, linear accumulation produces quadratic movement.

Knowing the denominators must follow a quadratic model $d_n = An^2 + Bn + C$, we can determine the coefficients directly by testing our first three known values ($n = 2, 3, 4$):

$$\begin{aligned} n = 2: \quad & A(2)^2 + B(2) + C = 4A + 2B + C = 2 \\ n = 3: \quad & A(3)^2 + B(3) + C = 9A + 3B + C = 6 \\ n = 4: \quad & A(4)^2 + B(4) + C = 16A + 4B + C = 12 \end{aligned}$$

Subtracting consecutive equations eliminates C :

$$\begin{aligned} (9A + 3B + C) - (4A + 2B + C) &\implies 5A + B = 4 \\ (16A + 4B + C) - (9A + 3B + C) &\implies 7A + B = 6 \end{aligned}$$

Subtracting these two yields $2A = 2$, meaning $A = 1$. Back-substituting gives $B = -1$ and $C = 0$.

The denominator sequence collapses into an exact closed form:

$$d_n = n^2 - n = n(n - 1).$$

Consequently, the dimensional ratio $r_n = D_n/D_{n-1}$ is determined for every step $n \geq 2$:

$$r_n = -\frac{1}{n(n-1)}.$$

With the recurrence relation $D_n = -\frac{D_{n-1}}{n(n-1)}$ established and the base case $D_1 = 1$, we can compute the first few terms directly:

$$\begin{aligned} D_1 &= 1 \\ D_2 &= -\frac{D_1}{2 \cdot 1} = -\frac{1}{2 \cdot 1} \\ D_3 &= -\frac{D_2}{3 \cdot 2} = \frac{1}{(3 \cdot 2) \cdot (2 \cdot 1)} \\ D_4 &= -\frac{D_3}{4 \cdot 3} = -\frac{1}{(4 \cdot 3 \cdot 2) \cdot (3 \cdot 2 \cdot 1)} \end{aligned}$$

Factoring the denominator of each successive term exposes how the recurrence accumulates:

$$D_n = D_1 \prod_{m=1}^{n-1} \left(-\frac{1}{m^2 + m} \right).$$

Because $m^2 + m = m(m + 1)$, we can split the denominator into two shifted products:

$$D_n = (-1)^{n-1} \cdot \frac{1}{\left(\prod_{m=1}^{n-1} m \right) \left(\prod_{m=1}^{n-1} (m + 1) \right)}.$$

The first product forms $(n - 1)!$, while the second shifts by one index to form $n!$:

$$\prod_{m=1}^{n-1} m = (n - 1)!, \quad \prod_{m=1}^{n-1} (m + 1) = 2 \cdot 3 \cdots n = n!.$$

Substituting these evaluations yields the official closed-form solution:

$$\det(A_n) = \frac{(-1)^{n-1}}{n!(n-1)!}.$$